Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/23/2004
PI Name:	Yamauchi, Mitsuo Ph.D., D.D.S.		
Project Title:	Mechanisms of Bone Loss Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	27599	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Duarte, Wagner Ph.D. (University of North Carolina)		
Grant/Contract No.:	NAG2-1596		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of this research is to understand, on the molecular, cellular and biochemical levels, the mechanisms of bone loss caused by the condition of microgravity. Our underlying hypothesis is that cells in weight-bearing bones, when subjected to microgravity/skeletal unloading, synthesize and deposit post-translationally altered type I collagen that does not mineralize properly leading to bone loss. During the last grant period, by employing a rat tail suspension model, we found that skeletal unloading resulted in accumulation of collagen with elevated lysine hydroxylation and altered cross-linking pattern in weight-bearing bones and that the alterations were bone-region specific. In addition, the lysyl hydroxylase gene expression was significantly up-regulated in these bones. In this renewal, as a natural extension of these findings, we now propose two major objectives: 1. to further define the skeletal unloading-induced collagen alterations in weight-bearing bones by employing a rat-tail suspension model, and 2. to investigate whether or not		

Task Description:	elevated lysine hydroxylation of collagen is a cause of impaired collagen and poor mineralization in vitro. The first aim will be achieved by biochemical analysis, newly developed bioimaging and molecular technologies that are complementary to one another. The second aim will be achieved by evaluating the effects of overexpression of lysyl hydroxylase genes in osteoblastic cell line on a type I collagen molecule, cross-linking and mineralization in vitro. These extensive and complementary studies will hopefully provide insights into the mechanisms of bone loss occurred during spaceflight and would help develop its preventive strategy. To understand the mechanisms of bone loss/defects induced by skeletal unloading/microgravity. This study may help to develop strategies to prevent the bone loss that occurs during spaceflight/unloading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the last grant period, by employing a rat tail suspension model, we found that skeletal unloading resulted in accumulation of collagen with elevated lysine (Lys) hydroxylation and altered cross-linking in weight-bearing bones (Shiiba et al, J. Bone Min Res 2002). In order to determine if these alterations of collagen are a cause of unloading/microgravity-induced bone defects, we generated several osteoblastic cell (MC3T3-E1)-derived clones that express higher levels of lysyl hydroxylase 2b (LH2b), an enzyme thought to be critical for Lys hydroxylation in the collagen telopeptides/cross-linking, and characterized the effects on collagen cross-linking, collagen fibril formation and matrix mineralization. The cell clones were cultured for 2 weeks, and the matrices produced were first analyzed for collagen cross-links. The results demonstrated that in these clones, the hydroxylysine-aldehyde (Hylald) derived cross-links were significantly increased whereas those of Lysald-derived were diminished in comparison with the controls. In contrast, when MC3T3-E1 cells were transfected with a LH2b antisense construct, the cross-links exhibited the opposite pattern, i.e. decreased Lysald-derived cross-links and increased Hylald derived cross-links. The data clearly demonstrate a critical role of LH2b in determining collagen cross-linking pathways. This is the first study demonstrating that the collagen cross-linking pathways (Lysald- vs. Hylald- derived) can be switched by manipulating the LH2b gene expression levels in osteoblastic cells. Furthermore, collagen fibrillogenesis and matrix mineralization in the clones expressing higher levels of LH2b were investigated. The diameter of collagen fibrils examined by transmission electron microscopy was markedly smaller in those clones than that of the controls. The onset of matrix mineralization in the clones was significantly delayed and considerably fewer mineralized nodules were formed in their cultures in comparison with the controls. When transplanted into immunodeficient mice, the clones failed to form mineralized matrices in vivo while a bone-like mineralized matrix was well formed by the controls. The diameter of the collagen fibrils and the timing/extent of matrix mineralization in vitro were inversely correlated with the level of LH2b. These results indicate that altered Lys hydroxylation/cross-linking of collagen may in part be a cause of unloading/ microgravity-induced bone defects.
Bibliography Type:	Description: (Last Updated: 08/11/2020)
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